

1. Same QA semantics, three output contracts

Question

What is the warranty period?

Context

[S1] The battery lasts 12 hours.

[S2] The device includes a 24-month warranty.

[S3] Accessories sold separately.

Plain

24 months

J

{"answer": "24 months"}

Q

ANSWER: 24 months

QUOTE: "The device includes a 24-month warranty."

2. Train one scaffold and direct overlay adapters

Frozen base LM
Qwen2.5-7B-Instruct

Scaffold **S**
plain-answer targets

Direct **F_J**
JSON targets

Direct **F_Q**
quote targets

3. Residualize in effective dense-delta space

Per adapted module m

$$\Delta_m = (\alpha_m / r_m) B_m A_m$$

$$\Gamma_{k,m} = \Delta_{F_k,m} - \beta_k \Delta_{S,m}$$

$$\Delta_{R_k,m} = \text{TSVD}_r(\Gamma_{k,m})$$

Validation selects β from $\{0.5, 0.75, 1.0, 1.25\}$
under the semantic-retention guardrail

Recovered residuals
R_J and **R_Q**

4. Compare recovery under the same strict contract

Prompt-only
 $M_0 + \Delta_S$

Direct
 $M_0 + \Delta_{F_k}$

SOAR
 $M_0 + \Delta_S + \Delta_{R_k}$

Parser-based metrics

- strict JSON schema validity
- exact quote match
- answer EM/F1

```
graph TD
    subgraph "1. Same QA semantics, three output contracts"
        Q[Question: What is the warranty period?]
        C[Context: [S1] The battery lasts 12 hours. [S2] The device includes a 24-month warranty. [S3] Accessories sold separately.]
        P[Plain: 24 months]
        J[J: {"answer": "24 months"}]
        Q2[Q: ANSWER: 24 months  
QUOTE: "The device includes a 24-month warranty."]
    end

    subgraph "2. Train one scaffold and direct overlay adapters"
        FLM[Frozen base LM: Qwen2.5-7B-Instruct]
        S[Scaffold S: plain-answer targets]
        FJ[Direct F_J: JSON targets]
        FQ[Direct F_Q: quote targets]
        FLM --> S
        FLM --> FJ
        FLM --> FQ
    end

    subgraph "3. Residualize in effective dense-delta space"
        M[Per adapted module m]
        Eq1["Delta_m = (alpha_m / r_m) B_m A_m"]
        Eq2["Gamma_k,m = Delta_Fk,m - beta_k Delta_S,m"]
        Eq3["Delta_Rk,m = TSVD_r(Gamma_k,m)"]
        V[Validation selects beta from {0.5, 0.75, 1.0, 1.25} under the semantic-retention guardrail]
        R[Recovered residuals R_J and R_Q]
        M --> Eq1
        M --> Eq2
        M --> Eq3
        Eq1 --> V
        Eq2 --> V
        Eq3 --> V
        V --> R
    end

    subgraph "4. Compare recovery under the same strict contract"
        PO[Prompt-only: M_0 + Delta_S]
        D[Direct: M_0 + Delta_Fk]
        SOAR[SOAR: M_0 + Delta_S + Delta_Rk]
        PBM[Parser-based metrics: strict JSON schema validity, exact quote match, answer EM/F1]
        PO --> PBM
        D --> PBM
        SOAR --> PBM
    end
```